

A Synthetic Cross-Sectional Momentum Study

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Abstract

This repository-owned document provides stable page evidence for Paper and future tree-builder contract tests. Its claims are intentionally synthetic and must not be interpreted as investment research.

1 Introduction

The introduction defines a twelve-month lookback signal.

The study asks how a simple cross-sectional ranking can be represented with explicit source pages and a reviewable hierarchy. Monthly returns are used only to make the fixture read like a small research paper. The section continues on the next page so its canonical span is 1-2.

Introduction continues on page 2.

1 Introduction (continued)

The continuation discusses turnover and transaction costs.

A realistic workflow must keep this page attached to the introduction even when a parser sees a new heading later on the same physical page. Turnover and costs are described qualitatively; no measured result is needed for the deterministic fixture contract.

2 Method

The method ranks securities by trailing excess return.

At each month end, the synthetic universe is ordered from the lowest to the highest trailing return. The procedure keeps missing values out of the ranking and records every decision before portfolio formation. Method details continue through pages 3 and 4.

2 Method (continued)

Signals are lagged by one period so that the illustrative construction does not use contemporaneous returns. Rebalancing occurs monthly and the same deterministic ordering resolves tied observations.

2.1 Portfolio Construction

The portfolio subsection forms equal-weighted quintiles.

Each quintile contains an equal number of eligible securities whenever the universe size permits. Residual observations are assigned in stable symbol order, making the example reproducible without a model call. The subsection continues on page 4 with the long-short definition.

This nested subsection establishes a three-level expected tree path.

2.1 Portfolio Construction (continued)

The top and bottom quintiles define the long-short portfolio.

The highest-ranked quintile is held long and the lowest-ranked quintile is held short. Gross exposure is normalized before any optional risk target is applied.

Risk scaling completes the method on the final page.

3 Limitations

The fixture uses synthetic observations and no investable returns.

The document omits live market data, parameter estimation, taxes, and capacity analysis. It exists only to exercise stable page anchors and tree topology in an offline test.

Method and limitations intentionally share page 4, so sibling spans overlap without violating the canonical Paper design contract.

This page closes the synthetic paper fixture.